

AI for Medical Imaging – Mid-project presentation

Incrementally Improving a Lightweight ENet for Thoracic CT Segmentation



23rd of September 2026

Students: Sakr Ismail, Steven Dong, Nordin el Assassi, Milan Tool

TA: Janmesh Ukey

Lecturer: Hoel Kervadec





Clinical task and objective

Task: 3D thoracic CT segmentation

- Input: a 3D chest CT scan.
- Output: one aligned 3D label mask.
- Classes: background, esophagus, heart, trachea, aorta.

Challenge

- Small/tubular organs occupy few voxels and have ambiguous boundaries.

citation

Why?

-
-

Goal

Improve the lightweight ENet incrementally & test whether segmentation improves in 3D.

Data pipeline and baseline



Baseline



Why extend?



citation

Preprocessing pipeline



Current preprocessing steps

1.

citation

Data exploration



🔍 Initial analysis

-



[maybe a figure here]

repr axial CT slice
w/ GT overlay

maybe: second panel w/ stitched
3D ground truth / prediction.

citation

Setup: fair comparisons and 3D evaluation



Fair comparisons

-

Current comparison

-

citation

Metrics per patient and organ

- **Dice**: region overlap $\uparrow$.
- **HD95**: robust surface distance in mm $\downarrow$.
- **ASSD**: mean symmetric surface distance $\downarrow$.
- Also logged: precision, recall, volume error, surface Dice.

Modification: bottleneck attention in ENet

? Hypothesis

Incremental variants

- **ENet**: unchanged architecture.
- **Channel attention**: weights feature channels.
- **Spatial attention**: weights deep spatial locations.
- **CBAM**: both; implemented, evaluation pending.



Same 2D input/output and training loop: a modular ENet
change, not a replacement framework.
citation

Preliminary 3D results: attention variants



Organ	Metric	ENet	Channel	Spatial	Best
Esophagus	Dice ↑	0.535	0.619	0.563	Channel
	HD95 mm ↓	27.52	19.84	19.63	Spatial
Heart	Dice ↑	0.795	0.824	0.830	Spatial
	HD95 mm ↓	18.70	26.29	15.81	Spatial
Trachea	Dice ↑	0.347	0.571	0.496	Channel
	HD95 mm ↓	96.20	54.51	37.29	Spatial

 Pattern

•

⚠ Preliminary: like placeholder to make extra clear this is preliminary doesn't have to be used obv

Internal 50-epoch preliminary validation results. Class 4/aorta omitted pending label/metric audit.

Interpretation



Observations



Verification required



Next steps



Steps

1.

Planned investigations

-

? Questions?

Take-away:

[1] CITATIONS